

AMERICAN UNIVERSITY OF PHNOM PENH

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American University of Phnom Penh

Final Project Progress Report

Topic: Khmer-text-to-speech

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ITM 454 002 - Natural Language Processing

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1. **Dataset Collection Status**

   We use the Khmer ASR Cultural Dataset (V2) from Mozilla Data Collective (https://mozilladatacollective.com/datasets/cml9h5vgc01bxmn075sjeftek), published by Digital Divide Data under a CC BY SA 4.0 license. The page lists 106.53 hours of speech and text pairs in WAV format, about 45.57 thousand instances, and native speakers from Cambodia. The publisher collected and labeled everything, and our metadata file confirms 45,572 sentences with no missing values. We selected one female speaker (f-adt2-0002) for a consistent voice, giving 11,377 sentences. After removing 4 rows with duplicated IDs and 752 sentences that were too short or too long or that contained Latin letters or digits, 10,621 remained. These were cleaned with Unicode normalization and removal of zero width spaces, newlines, and quotation marks. The audio was 48 kHz mono, totaling 22.09 hours. We checked quality, rejected 211 clips (98 percent passed), and converted the rest to 22,050 Hz with silence trimmed and loudness normalized. The final 10,410 clips (21.38 hours) were split by paragraph into 9,616 training clips (19.66 hours), 395 validation clips (0.82 hours), and 399 test clips (0.90 hours).

   Dataset distribution

   The data is clearly imbalanced by topic. Across all speakers, Recipes has 17,028 of the 45,572 sentences (about 37 percent), while Public Holiday has only 1,492 and Local Name 1,648. The same pattern holds in our training set, where Recipes has 3,449 clips and Public Holiday has 185. The metadata shows 11 topics and 347 subtopics. Speakers are also uneven. Three speakers read the same script of 11,377 sentences each, three others have between 2,345 and 5,571, and four have only 16. Since Khmer is written without spaces between words, we measured sentence length in characters: the mean is about 101, the median is 96, and the range is 21 to 381. Clips average 7.49 seconds with a standard deviation of 2.49 seconds.

2. **Training Done**
   - Fine-tuned a single-speaker VITS model for Khmer TTS (Coqui TTS, Colab GPU), starting from the pretrained English LJSpeech VITS.
   - Trained on 9,180 clips (about 18 h, 22050 Hz) and validated on 372, with batch size 16 and learning rate 2e-4.
   - Early stopping on validation mel loss ended training at epoch 16 of 50. The best loss was about 16.22.
   - Generated 14 sample waves (in-domain and out-of-domain) for later evaluation.

3. **Next Steps**

   For the remaining tasks, our next steps include:
   - Resume and finish/complete training (by 6 Nov)
   - Objective evaluation metrics (by 13 Nov)
   - Subjective evaluation (by 18 Nov)
   - Error analysis and fixes (by 20 Nov)
   - Final report and presentation (by 26 Nov)

References

Eren, G., & The Coqui TTS Team. (2021). *Coqui TTS* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.6334862

Ito, K., & Johnson, L. (2017). *The LJ Speech Dataset*. https://keithito.com/LJ-Speech-Dataset/